Disease Prediction Using Ensemble Learning

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***Abstract*- In response to evolving health information needs globally, our project employs a combination of advanced machine learning algorithms—Decision Tree, Random Forest, and Gradient Boosting—to predict diseases accurately based on user-input symptoms. Targeting an accuracy range of 92-95%, this predictive system has the potential to revolutionize healthcare by enabling early disease prediction. By facilitating proactive preventive measures, the model aligns with the adage "Prevention is better than cure." Our supervised machine learning approach empowers individuals to assess their health, make informed decisions, and reduce unnecessary pressure on hospitals. The system's efficiency lies in its ability to predict diseases, offering a valuable tool for users while contributing to smarter healthcare solutions. The final model used is the Voting Classifier that combines 3 separate models. It exhibits an accuracy of 95%. The User Interface (UI) of our system has been meticulously designed to provide a seamless and user-friendly experience to individuals seeking health insights. The interface is intuitively crafted, ensuring that users can effortlessly navigate through the system. The design prioritizes clarity and simplicity, allowing users to input their symptoms easily and interpret the results without any complexity.**

Keywords—Decision Tree, Random Forest, Usability, Gradient Boosting, Voting Classifier

I. INTRODUCTION

Disease prediction is of vast importance in today's world as it provides options to healthcare officials about a patient's future state. Disease prediction is essential in healthcare for several reasons. Firstly, it allows for the early identification of health issues before they become severe. When diseases are predicted early, treatments can be more effective and less invasive, leading to better outcomes for patients. Additionally, it enables healthcare providers to offer personalized preventive measures based on an individual's risk factors, lifestyle, and medical history. This proactive approach can help people make lifestyle changes or take precautions to reduce their chances of developing certain illnesses. Disease prediction also plays a significant role in resource management within healthcare systems. By anticipating disease prevalence and outbreaks, healthcare resources such as facilities, staff, and medications can be allocated more efficiently, ensuring that adequate care is available when needed. Ultimately, disease prediction contributes to reducing healthcare costs by minimizing the need for expensive treatments that could have been prevented or managed earlier.

Our project revolves around leveraging sophisticated computer techniques to predict various diseases comprehensively. We're exploring a broad spectrum of health conditions to gain a deeper understanding and more accurate predictions. Our approach involves integrating three robust methodologies – Decision Tree, Random Forest, and Gradient Boosting – into a unified system to create a highly reliable predictive model.

However, our aspirations exceed merely early disease detection. We envision our project as a potential game-changer in healthcare, transforming how medical professionals tailor treatments. By harnessing the analytical power of artificial intelligence to scrutinize individual patient data, we aim to optimize treatment plans. This could significantly elevate patient outcomes and, intriguingly, might even drive down healthcare expenditures.

Central to our endeavor is a steadfast commitment to ethical standards and stringent data privacy measures. We prioritize adhering to ethical guidelines and implementing robust safeguards to ensure the confidentiality and integrity of sensitive information while pushing the boundaries of innovation in healthcare.

Our primary objective is to engineer an intelligent diagnostic system capable of foreseeing illnesses based on symptoms provided by users. This initiative seeks to alleviate the strain on hospital outpatient departments (OPDs), lightening the workload for medical professionals while offering users a convenient avenue for initial health assessments. To accomplish this, we're planning to deploy three distinct algorithms, bolstering the dependability and precision of our predictive model.

In addition to its predictive prowess, our system will boast a seamless, user-friendly interface. We understand the significance of continuously gathering data for refinement. Thus, our system will encompass a database to house user-entered information, aiding in the evolution of future versions.

Beyond mere functionality, our project prioritizes user comprehension. We'll integrate diverse visual representations of both collected data and model outcomes, fostering a lucid understanding of the diagnostic process. By amalgamating these features, our aspiration is to craft a sophisticated yet accessible healthcare tool. This tool won't just serve its immediate purpose; it'll also lay the groundwork for ongoing advancements in predictive healthcare technology.

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II. LITERATURE SURVEY

The study mentioned in **[1]** explores healthcare data mining's effectiveness in predicting heart disease in individuals with diabetes. Despite available data mining classification algorithms for heart disease, a gap exists in diabetic cases. The research emphasizes fine-tuning the decision tree model, consistently superior to naive Bayes and support vector machine models, for optimal forecasting of heart disease likelihood in diabetic individuals. This highlights the importance of refining predictive models, particularly the decision tree, in addressing diabetes-related heart disease. Essentially, decision tree seems to perform well for classification problems.

According to the study in **[2]** which explores the role of deep learning algorithms in disease prediction, focusing on Artificial Neural Network (ANN), FM-Deep Learning, Convolutional Neural Network (CNN), and Recurrent Neural Network (RNN). The paper addresses the current challenges in disease prediction, such as poor interpretability and data quality issues, providing temporary solutions. The conclusion reviews structured and unstructured data algorithms, highlighting their applications and the challenges they face. Despite being mainstream, these algorithms encounter issues like sample imbalance and poor interpretability. Deep Learning techniques are excellent at making predictions for datasets with multiple features but the main issue resides in their inability to perform well with smaller datasets. This is where classifiers hold an advantage.

This study **[3]** addresses the critical tasks of diagnosing and prognosing cardiovascular disease, emphasizing the importance of accurate classification for effective treatment. Utilizing machine learning to recognize patterns and enhance diagnostic precision, the research introduces a k-modes clustering method with Huang starting to improve classification accuracy. Employing models such as random forest, decision tree classifier, multilayer perceptron, and XGBoost, the proposed approach underwent parameter optimization using GridSearchCV. Applied to a Kaggle dataset of 70,000 instances, the models achieved notable accuracies, with the multilayer perceptron exhibiting the highest accuracy of 87.28%, outperforming other algorithms. The study underscores the potential of machine learning, particularly the multilayer perceptron, in predicting and reducing the fatality associated with cardiovascular diseases.

The study mentioned in **[4]**, explores the application of Machine Learning (ML) in healthcare, specifically focusing on heart disease and locomotor disorders using a UCI dataset. Employing supervised learning methods such as Naive Bayes, SVM, Logistic regression, Decision Tree Classifier, Random Forest, and K-Nearest Neighbor, the study demonstrates the effectiveness of KNN with eight neighbors in terms of precision, sensitivity, accuracy, and F1-score compared to other methods. The conclusion highlights that ML methods outperform statistical techniques in predicting and classifying heart disease, supporting previous studies. The article suggests future work could include emerging classification models for complex time series datasets and extend the research to other biological diseases.

This study **[5]** , focuses on the early prediction of stroke using machine learning (ML) models. The study introduces a stacking method, employing naive Bayes, random forests, RepTree, and J48 as base classifiers, with a logistic regression meta-classifier. The stacking classification demonstrates superior performance with an AUC of 98.9%, F-measure, precision, and recall of 97.4%, and an accuracy of 98%, outperforming other methods. The conclusion emphasizes the potential of ML in early stroke prediction, enabling clinicians to identify relevant risk factors and assess the probability of stroke occurrence. The study suggests future work involving the enhancement of the ML framework and the exploration of image data from brain CT scans for predictive modeling.

This particular academic paper **[6]** delves into an in-depth exploration of Coronary Artery Disease (CAD) and the Machine Learning techniques associated with its detection. The study emphasizes utilizing the synthetic minority oversampling technique as a part of its initial data processing. Subsequently, the authors employ various Machine Learning models like Naive Bayes, logistic regression, J48, random forest, rotation forest, multilayer perceptron, and ANN. The effectiveness of these models is assessed by comparing their performance with and without the incorporation of SMOTE, employing a 10-fold cross-validation method. Moreover, ensemble techniques such as bagging, stacking, and voting classifier are applied to augment the results. The evaluation of the models is conducted using standard metrics including precision, recall, accuracy, and AUC. The research highlights the stacking ensemble model post-SMOTE with 10-fold cross-validation as the most superior, achieving an accuracy of 90.9%, precision of 96.7%, recall of 87.6%, and an AUC of 96.1%. These findings represent the central proposition of the paper.

The following paper **[7]** deals with the prediction of Parkinson's disease. In the beginning, the authors highlight the seriousness of this disease by displaying staggering number of people succumbing to the disease. Early prediction and preventive measures are crucial to curbing these fatalities. Ther proposed method in this paper demonstrates superior performance compared to established machine learning algorithms commonly used for predicting cardiac issues. Through a comprehensive evaluation that included Decision Tree, SVM, KNN, Random Forest, and XG-Boost, the method consistently outperformed these approaches in accuracy, MCC, and f1score. Further investigation strongly indicates the effectiveness of the approach over existing methods. Specifically, the ensemble model technique which is a voting classifier showcased superior performance compared to other machine learning models, exhibiting an impressive accuracy of 94.87%, an MCC of 81.99%, and an score of 94.52% according to the authors' thorough analysis.

This research paper **[8]** deals with the detection of chronic kidney disease(CKD). This research introduced several machine learning algorithms aimed at diagnosing Chronic Kidney Disease (CKD) in its early stages. The models created using data from CKD patients underwent training and validation using the previously mentioned input parameters. Research was conducted to understand the relationships between various factors, allowing for the reduction of features and the removal of redundant information. Distinct sets of outputs were presented, each

stemming from various parameter values assigned to individual models. Notably, XgBoost exhibited the highest performance, achieving an accuracy of 0.9833 with the original CKD dataset. After implementing PCA, this accuracy surged to 0.9916 for the same classifier. Conversely, AdaBoost, random forest, gradient boosting, LGBM, and Extra Tree maintained an accuracy of 0.9833, consistent with the original CKD dataset. Remarkably, the dataset's highest accuracy of 0.9833 improved to 0.9916 upon applying PCA specifically to the XgBoost classifier. However, KNN and MLP displayed comparatively poor performance. The ANN classifiers showed a 60% accuracy for both datasets, primarily due to limited data availability. A comprehensive table was compiled, encompassing experimental results for each model, detailing their testing and training accuracy, F1-measure, precision, recall, and confusion matrix.\

This research paper **[9]** discusses about the severity of liver disease and how to apply ML methods and techniques to detect it in humans. Liver disease poses a grave threat to human life, necessitating immediate medical attention. Traditionally, healthcare professionals have relied on pathological approaches to assess a patient's condition. This study aimed to use machine learning techniques for early liver disease prediction. Multiple machine learning models—such as NB, SVM, LR, ANN, kNN, J48, RF, RT, RepTree, RotF, AdaBoostM1, Stacking, Bagging, and Voting—were assessed based on metrics like Accuracy, Precision, Recall, F-Measure, and AUC to anticipate liver disease occurrence. The findings revealed that the Voting classification method surpassed others, achieving an Accuracy, Recall, and F-measure of 80.1%, Precision of 80.4%, and an AUC of 88.4% after employing SMOTE with 10-fold cross-validation. Consequently, the Voting method emerged as the primary focus of this study. Notably, this particular proposed model (Voting) demonstrated superior accuracy compared to similar research papers using the same dataset features as in this study.

This Paper[**10**]In 2016, human-generated data exceeded ten exabytes. Exploratory Data Analysis (EDA) assesses data using graphical or non-graphical, and univariate or multivariate methods. Diagnostic methods for diseases are invasive or non-invasive, with non-invasive machine learning algorithms like SVM, K-means clustering, KNN, ANN, Naive Bayes, Logistic Regression, and rough set.Predicting heart disease involves factors like physical examination, symptoms, and influencing elements such as cholesterol, smoking, obesity, family history, blood pressure, and work environment. Machine learning, crucial in accurate prediction, leverages technology to handle large, unstructured data. This paper proposes K-means clustering in a big data environment, visualized using a Tableau dashboard.

This Paper [**11**]Processing raw healthcare data for heart information is crucial for early detection of heart abnormalities and saving lives. This study utilizes machine learning, specifically the hybrid HRFLM approach (Random Forest and Linear Method), to accurately predict heart disease. Early detection is vital for reducing mortality rates, and future research should focus on real-world datasets and explore various machine learning techniques for enhanced prediction. Developing new feature-selection methods is also recommended to improve the performance of heart disease prediction.

This Paper [**12**]Creating Online Augmented Reality instructional media for the third-year Mycology course (317216) on Phylum Basidiomycota at the Faculty of Science, Khon Kaen University enhances its accessibility and usability. Student satisfaction, with an average score of 4.47 out of 5, reflects a highly positive response. The key features include easy accessibility within an online system. However, limitations exist, as the system is currently compatible only with Android devices (version 2.2 and above). Future developments could focus on expanding compatibility to iOS and Windows Phone, as well as diversifying content for a broader range of instructional media.

This paper **[13]** delves into the critical matter of combating dengue by presenting a comprehensive approach utilizing Machine Learning algorithms for disease detection. The authors meticulously begin by scrutinizing the performance of distinct models such as logistic regression, decision tree, SVM, K-NN classifier, and Naive Bayes classifier individually. Subsequently, they introduce and thoroughly evaluate a hybrid ensemble machine learning model, meticulously analyzing its efficacy in dengue disease prediction. Upon thorough analysis, the authors conclusively establish that the hybrid ensemble technique significantly enhances accuracy levels. This highlights the pivotal role and substantial promise of employing hybrid ensemble methodologies within the expansive domain of Machine Learning, specifically in disease detection and medical diagnostics.

III. METHODOLOGY

A. Machine Learning Models Used

1) Decision Tree:

A Decision Tree is a versatile supervised machine learning algorithm used for classification and regression tasks. It creates a tree-like structure by recursively splitting data based on the most significant features.

At each decision node, the algorithm selects the feature that maximizes information gain (for classification) or minimizes mean squared error (for regression). Information Gain measures the reduction in entropy, denoted as

$$Entropy(S) = -p_1 log_2(p_1) - p_2 log_2(p_2)$$

for a binary classification problem. The algorithm recursively applies these steps to create a tree until a stopping condition is met, such as a predefined depth or a node containing examples of only one class.

The resulting tree structure provides decision rules derived from the paths taken from the root to the leaves. These rules guide the classification or regression process.

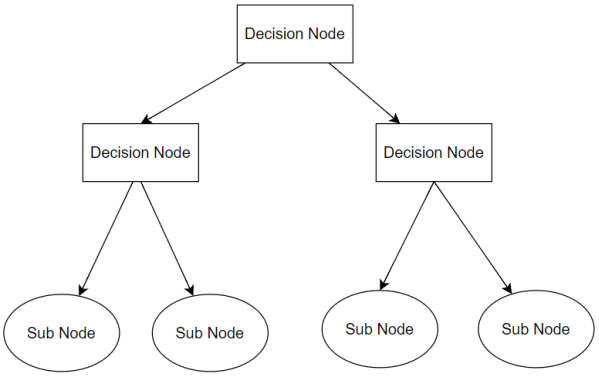


Fig. 1. The figure depicts the working of a Decision Tree

2) Random Forest:

Random Forest is an ensemble learning algorithm that operates by constructing a multitude of decision trees during training and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees.

It operates by employing bootstrapped sampling, where each tree is trained on a different subset of the dataset drawn with replacement. Additionally, feature randomization is introduced during tree construction, ensuring that only a random subset of features is considered at each split. This diversity prevents overfitting and enhances the model's robustness.

Each decision tree is grown until a specified stopping condition is met, such as reaching a certain depth. The final prediction for a given instance is determined through a majority vote (for classification) or averaging (for regression) of predictions from all individual trees. Random Forests excel in improving generalization performance, handling complex datasets, and mitigating overfitting.

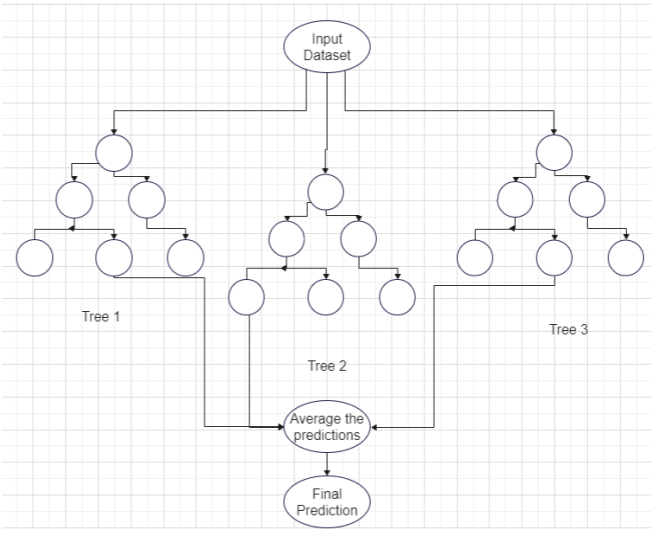


Fig. 2. Random Forest Classification

3) Gradient Boosting:

Gradient Boosting is an ensemble learning technique that sequentially builds a robust predictive model by combining the outputs of weak learners, often decision trees.

Unlike independent tree construction in methods like Random Forests, Gradient Boosting develops trees sequentially, each addressing the errors of the combined model from previous trees.

Initially, a simple model, typically a shallow decision tree, is created for the first prediction. Subsequently, the difference between actual and predicted values (residuals) is calculated, and a new tree is constructed to predict these residuals. This process iterates, with each new tree focusing on minimizing errors from the ensemble of existing trees.

The learning rate parameter controls the contribution of each tree, and the final prediction is the sum of predictions from all individual trees. This sequential, adaptive learning approach enhances the model's accuracy, making Gradient Boosting a powerful ensemble method widely used in machine learning, with implementations such as AdaBoost and XGBoost.

4) Voting Classifier:

A Voting Classifier is an ensemble learning technique that combines the predictions of multiple base models to make a final decision. This method aggregates the results from different classifiers, each of which may use diverse algorithms or configurations.

In a typical setup, a majority vote or a weighted vote is used to determine the final prediction. This approach leverages the diversity among individual classifiers, allowing the ensemble to benefit from the strengths of each. Voting classifiers can be employed for both classification and regression tasks, making them versatile tools in machine learning.

Common types of voting classifiers include Hard Voting, where the majority prediction is selected, and Soft Voting, which considers the confidence scores of individual classifiers and assigns weights accordingly. This ensemble method often enhances overall predictive performance and robustness, particularly when individual models may have different strengths or weaknesses on specific aspects of the data.

B. *The combined Machine Learning Process*

We utilized an the voting class ensemble method to combine the predictions made by the Decision Tree Classifier, Random Forest classifier, and the Gradient Boosting Algorithm.

A Voting Classifier with hard voting combines the predictions of individual classifiers, namely Decision Tree, Random Forest, and Gradient Boosting, to make a collective decision. Each of these slow learners contributes its prediction, and the majority vote determines the final output.

The Decision Tree, as a base classifier, makes predictions based on a set of rules learned from the training data. It is a simple yet effective model that captures patterns in the data, but it may have limitations in handling complex relationships or noisy data.

The Random Forest, another base model, builds multiple decision trees through bootstrapped sampling and feature randomization. By aggregating the predictions of these trees, the Random Forest mitigates overfitting and enhances overall predictive accuracy.

Gradient Boosting, the third base model, constructs decision trees sequentially, with each tree focusing on correcting the errors of the combined ensemble. This iterative learning process allows Gradient Boosting to adapt and improve over multiple iterations, producing a powerful predictive model.

The Voting Classifier combines these diverse models by allowing each to contribute its prediction, and the majority vote determines the final output. In the case of hard voting, the class with the most votes becomes the ensemble's prediction. This approach leverages the strengths of individual models, compensating for potential weaknesses in one with the strengths of others.

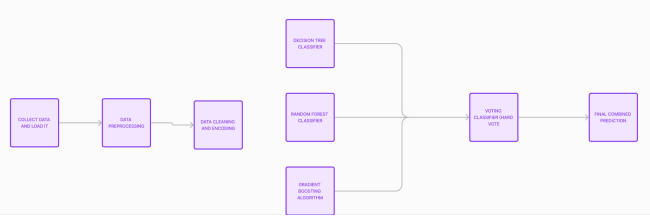


Fig. 3. The proposed Machine Learning Model

C. *Process Model Used*

In adopting an iterative and incremental development process model for our disease prediction system, we embark on a structured journey encompassing distinct phases aimed at refining and enhancing the system's capabilities over time.

The process commences with a thorough engagement with stakeholders to identify and document the objectives of our disease prediction system. This collaborative effort ensures a clear understanding of the goals, setting the foundation for subsequent development phases.

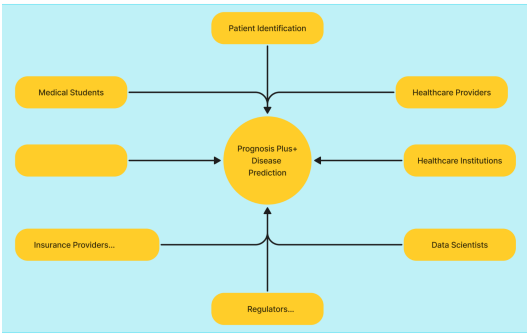


Fig. 4. Stakeholder Identification

Following the definition of objectives, a comprehensive system design is crafted, emphasizing scalability and flexibility. This step lays the groundwork for a robust architecture that accommodates evolving requirements and facilitates seamless integration of machine learning algorithms.

A critical aspect of the iterative and incremental approach involves selecting suitable machine learning algorithms, such as Decision Tree, Random Forest, and Gradient Boosting. The process includes meticulous parameter tuning to optimize the algorithms for accurate and efficient disease prediction.

Gathering relevant datasets and preparing the data for training constitute the subsequent step. Rigorous data preprocessing ensures the quality and appropriateness of the information used to train and refine our predictive models. Through label encoding and cleaning the rest of the data, an assurity towards a highly accurate model gets established.

Implementation of the selected algorithms and training of models take center stage in this phase. Additionally, the incorporation of a Voting Classifier for ensemble predictions enhances the overall predictive power of the system.

Simultaneously, an intuitive and user-friendly interface is designed to facilitate seamless interaction. This step acknowledges the importance of user experience in ensuring effective symptom input and result interpretation.

Rigorous testing is conducted across various dimensions, encompassing functionality, accuracy,

and usability. This iterative testing process identifies potential issues and refines the system's performance before deployment.

Upon successful testing, the system is deployed to production, marking a pivotal transition from development to practical application. The deployment phase involves careful consideration of factors such as scalability and accessibility.

User feedback becomes a crucial input for the iterative aspect of the development model. Continuous refinement of predictive models and the user interface is guided by user experiences and evolving requirements.

As the system becomes operational, ongoing maintenance measures are implemented to address emerging issues and ensure continuous effectiveness. Scalability measures are incorporated to accommodate potential increases in data volume and user interactions.

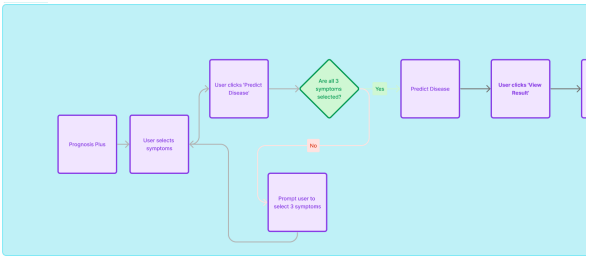


Fig. 5. Block Diagram of the application

D. *User Interface Framework Walkthrough*

In navigating through the user interface (UI) framework for our website's disease prediction feature, users are guided through a seamless and intuitive process. The walkthrough encompasses the following steps:

1) Accessing Disease Prediction:
Users initiate the process by clicking on the "Disease Prediction" section, prominently available on the website's main interface. This serves as the entry point for individuals seeking an accurate prediction of potential health conditions

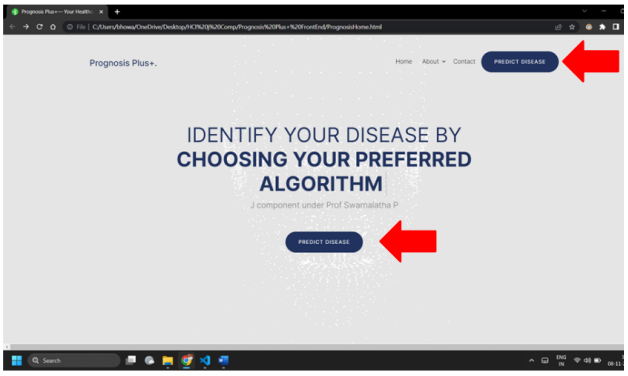


Fig. 6. Depiction of Prediction Button

2) Symptom Selection:
Subsequently, users are prompted to engage in a straightforward symptom selection process. They are encouraged to select symptoms from a comprehensive list, beginning with the first symptom of their choice. The UI framework ensures ease of use, providing a clear and visually accessible set of symptoms.
3) Sequential Symptom Selection:
The UI guides users through the sequential selection of symptoms, prompting them to choose the second and third symptoms from the available options. This step-by-step approach streamlines the input process, allowing users to articulate their symptoms accurately.

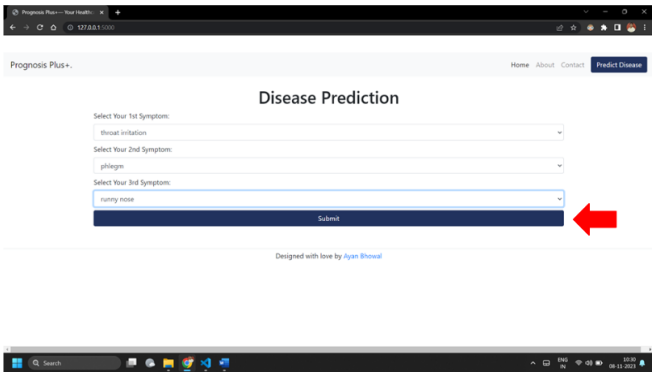


Fig. 7. Depiction of selecting the disease

4) Initiating Disease Prediction:
After the symptom selection phase, users are prompted to proceed by clicking on the "Predict Disease" button. This action signals the system to initiate the disease prediction algorithm based on the provided symptoms.
5) Display of Predicted Disease:
Upon completion of the prediction process, the UI framework dynamically transitions to a dedicated page displaying the accurately predicted disease. This page is designed to present the results in a comprehensible and visually appealing manner, ensuring users can easily interpret and understand the outcome.

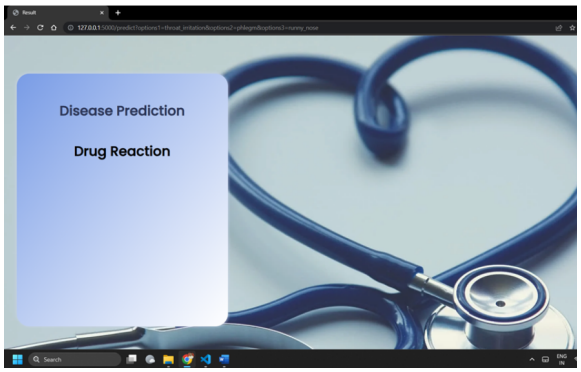


Fig. 8. Displaying the predicted disease

The UI framework is crafted to prioritize user experience, offering an intuitive and straightforward journey through the disease prediction feature. The step-by-step symptom selection process aims to simplify user input, while the clear presentation of the predicted disease enhances the overall usability of the system. The framework's design aligns with the goal of providing users with a reliable and user-friendly tool for obtaining accurate predictions related to their health conditions.

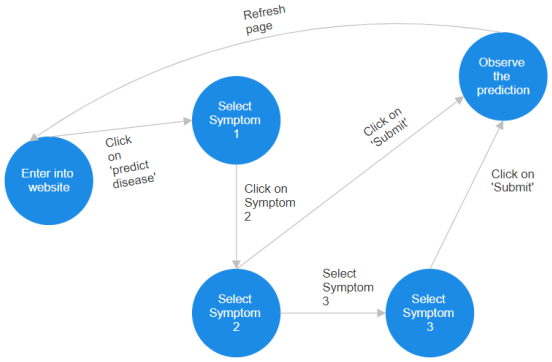


Fig. 9. State Transition Network of the Interface

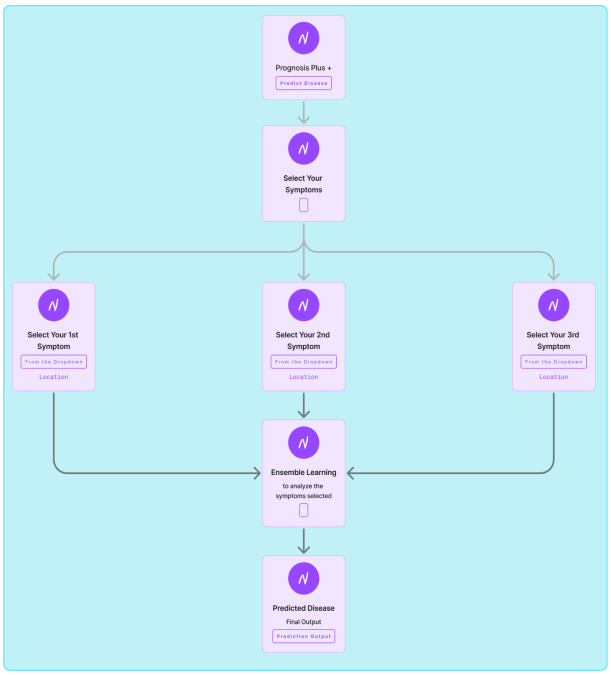


Fig. 10. Hierarchical Task Analysis

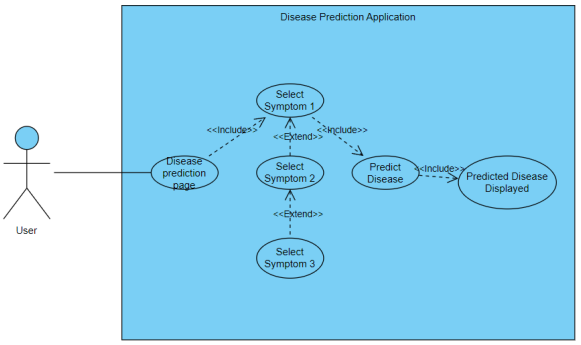


Fig. 11. Use Case Diagram

The use case diagram for our project serves as a visual representation of the various interactions and functionalities within the disease prediction system. It captures the primary actors, including users and the system itself, along with the essential use cases that depict the distinct functionalities.

Actors, such as website visitors or individuals seeking disease predictions, interact with the system through key use cases like accessing disease prediction, symptom selection, and initiating the prediction process. This diagram provides a high-level overview of the user-system interactions, offering a clear understanding of the system's functionality and the roles different actors play in the disease prediction workflow.

The use case diagram serves as a valuable tool for project stakeholders and developers to conceptualize and communicate the core functionalities and user interactions within the disease prediction system.

IV. RESULTS AND DISCUSSION

A. Results

TABLE I. Evaluation for the model

Accuracy	Precision	Recall	F1 Score
0.951219 51219 51219	0.926829 26829 2683	0.951219 51219 51219	0.934959 34959 34958

1) *Accuracy (95.1%):*
Accuracy measures the overall correctness of our model.
Our model correctly predicts the class for approximately 95.12% of the instances in the test set. High accuracy is generally good, but it might not be the only metric to consider, especially in imbalanced datasets.

2) *Precision (92.7%):*
Precision measures our model's ability not to label as positive a sample that is negative.
In our case, the model correctly identifies positive instances approximately 92.68% of the time.
A high precision indicates that when the model predicts a positive class, it is likely correct.

3) *Recall (95.1%):*
Recall (or sensitivity) measures our model's ability to capture all positive instances.
Our model captures approximately 95.12% of the actual positive instances.
A high recall indicates that the model is effective in identifying most of the positive cases.

4) *F1 Score (93.5%):*
The F1 score is the harmonic mean of precision and recall. It balances false positives and false negatives.
Our F1 score of approximately 93.50% suggests a good balance between precision and recall.
This is particularly useful when we want to avoid either too many false positives or too many false negatives.

B. GOMS

GOMS Table

Task	Operators	Subtasks
1. Navigate to Website	P: Recognize, K: Know, M: Use, S: -	- Use the web browser (P, K). - Type the Prognosis Plus+ website URL in the address bar (M).
2. Homepage Interaction	P: Observe, K: Know, M: Click, S: Use verbal feedback	- Observe and recognize homepage call-to-action buttons (P). - Choose an action based on preference (K, S). - Click the selected button using the mouse (M).
3. Redirect to Form	P: Understand, K: Know, M: Follow	- Understand the redirection process (K). - Follow the redirection to the form page (M).

C. Schniedermann's 8 golden rules

Strive for consistency	The utilization of icons and consistent color schemes within the website and its user interface significantly enhances the overall user experience (UX).
Enable frequent users to use shortcuts	Incorporated multiple call-to-action buttons to facilitate users in efficiently achieving their objectives within the user experience cycle.
Offer informative feedback	Upon completion, a subsequent page unfolds, furnishing the predicted disease as insightful feedback.
Design dialogs to yield closure	The conclusive page provides a clear and comprehensive description, affirming the successful prediction of the disease through precise and informative text. The landing page, adeptly designed, delivers a lucid explanation of the website's purpose and user-friendly usage guidelines.
Offer simple error handling	Users will receive prompts in case of errors while selecting symptoms from the dropdown menu, ensuring a smooth and error-free interaction.
Permit easy reversal of actions	Incorporated a "Go back" button to facilitate a seamless reversal of actions, enhancing user flexibility and navigation.
Support internal locus of control	The users can freely navigate through the website and their data will stay only visible to them. The predicted disease is only shown on the basis of the user's choice of symptoms.
Reduce short-term memory load	The symptoms are chosen by the user from a list rather than having them recall and type it themselves.

D. Confusion matrix

The presence of a diagonal of 1s in our confusion matrix is a positive indication of our model's

performance. Each element along the diagonal represents the number of instances that were correctly classified for each class, or in other words, the true positives (TP).

This pattern suggests that our model is making accurate predictions across various classes, demonstrating its ability to correctly identify and distinguish between different diseases or conditions. The higher the values along this diagonal, the more instances our model has successfully classified.

In summary, a strong diagonal presence in the confusion matrix signifies that our model is performing well in terms of correctly predicting the classes across the different categories, showcasing a robust and accurate classification capability.

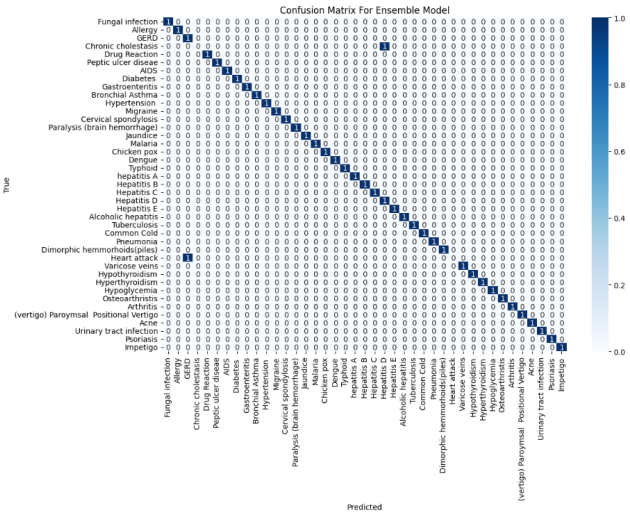


Fig. 12. This is the confusion matrix that has been generated from the ensemble model

E. Further Discussion

The decision to incorporate Decision Tree, Random Forest, and Gradient Boosting in the ensemble was based on the desire to harness the unique strengths of each model. Decision Tree offers simplicity and interpretability, Random Forest provides robustness against overfitting, and Gradient Boosting adapts and improves iteratively. By combining these models through hard voting, we sought to create an ensemble that compensates for individual model weaknesses and leverages their collective predictive power.

Our analysis revealed that the ensemble model with the Voting Classifier consistently demonstrated competitive performance compared to individual models. The decision-making diversity inherent in the ensemble allowed it to capture a broader range of patterns within the dataset. Notably, the ensemble's ability to achieve accurate predictions without overfitting, as seen in the Random Forest, and adaptability, akin to Gradient Boosting, contribute to its robust performance.

The ensemble model strikes a balance between interpretability and complexity. While Decision Tree provides an interpretable decision-making process, the ensemble's aggregation of multiple models introduces complexity. However, the trade-off is justified by the superior predictive performance achieved by the ensemble.

Despite the success of the ensemble model, it is crucial to acknowledge potential challenges. The choice of ensemble composition and hyperparameter tuning may impact results, and further investigation is warranted to optimize these aspects. Additionally, the exploration of unsupervised learning techniques could enhance the model's capacity to uncover latent patterns in the data, particularly as the dataset expands.

INTERFACE TESTING

1. Interface testing in human-computer interaction (HCI) refers to the evaluation and examination of the user interface of a software application, system, or device. This testing aims to assess various aspects of the interface's usability, functionality, and user experience. It involves scrutinizing how well the interface meets the needs of its intended users and ensures smooth interaction.

2. The interface holds significant importance in captivating customers' interest in a specific product or service. Therefore, in our project, we ensured that the interface we developed aimed to grab the user's attention.

3. An approach to accomplish this is by crafting a visually appealing design. That's why we incorporated calming color schemes to enhance the interface's aesthetics.

4. Testing the interface's ability to handle errors or unexpected user inputs gracefully without crashing or causing confusion is also an important functionality.

5. Furthermore, the user receives their diagnosis almost instantly, ensuring a favorable response time for the user's convenience. This means that the system or service provides the user with their diagnosis or information extremely quickly, making sure that the time taken for the system to respond is very short, thereby enhancing the user experience. A quick response time is essential as it contributes to a smoother and more efficient interaction between the user and the system, preventing frustration or delays in obtaining important information or results.6. Validation of Business Processes: It verifies that the software

aligns with the business processes and goals it was designed for.

6. Bug Reporting and Resolution: Any issues, bugs, or discrepancies found during UAT are reported, tracked, and addressed before the final release.

User Testing

1. User Acceptance Testing (UAT) is a critical phase in the software development lifecycle where the focus shifts to ensuring that the developed software or system meets the requirements and expectations of the end users. It's the final testing phase before the product is released to the users or customers. The primary goal of UAT is to validate the system's readiness for real-world use and to gain confidence that it satisfies user needs. Some of the key aspects of UAT are:-
2. End-User Involvement: UAT involves actual users or representatives of the end-users who test the software in a realistic environment or scenario.
3. Requirement Verification: UAT ensures that the software meets the specified requirements and functions as intended, addressing user needs effectively.
4. Real-World Scenarios: The testing process aims to simulate real-world scenarios, workflows, and use cases that users are likely to encounter.
5. Feedback Collection: Users provide feedback on usability, functionality, performance, and any deviations from their expectations.

V. CONCLUSION

In conclusion, our disease prediction system, incorporating ensemble learning with Decision Tree, Random Forest, and Gradient Boosting algorithms, represents a significant advancement in healthcare technology. Through an iterative and user-centric development process, we have successfully created a predictive model that not only achieves high accuracy in disease prediction but also offers a seamless and intuitive user interface.

Through rigorous testing and iterative refinement, we have enhanced the performance of our system. Fine-tuning algorithm parameters, optimizing data preprocessing, and constantly revising the usability has contributed to the system's accuracy and reliability. The ensemble approach, combining the strengths of multiple algorithms, has proven to be a robust solution, providing a more comprehensive and accurate disease prediction capability.

One of the key strengths of our system lies in its applicability in real-time environments. The efficiency of the algorithms, coupled with the user-friendly interface, allows for quick and reliable predictions. In a real-time setting, our system can aid healthcare professionals and individuals alike in making prompt decisions regarding disease prevention and management.

To further elevate the system's performance, future enhancements could explore additional algorithms, continuous learning mechanisms, and incorporation of emerging medical data. Machine learning models, by nature, can benefit from ongoing updates and adaptations to stay current with evolving medical knowledge and symptom patterns. Hence, testing the data with stronger models, unsupervised learning, neural networks etc, can raise the efficiency and ease of usability.

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